



DEPARTMENT OF MATHEMATICS
FEDERAL UNIVERSITY OF LAFIA

B.Sc. DEGREE FIRST SEMESTER ASSIGNMENT 2022/2023 Session

Course Code: MTH 111 Course Title: General Mathematics I

Unit:3

1. If α and β are the roots of the equation $3x^2 - 5x + 9 = 0$, find the value of $\alpha^3 + \beta^3$.
2. A progression has a first term of 12 and a fifth term of 8. Find the sum of the first 25 terms if the progression is geometric.
3. A geometric progression has first term 64 and sum to infinity 256. Find the common ratio and the sum of the first 10 terms.
4. Expand $(5x - 4)^{10}$
5. Find the coefficient of z^4 in the expansion of $(5 + z)^{20}$
6. Given $A = \begin{pmatrix} 7 & -3 \\ 4 & -3 \end{pmatrix}$ and $B = \begin{pmatrix} -2 & 1 \\ -3 & 2 \end{pmatrix}$. Find AB^{-1} .
7. What value must x have, so the matrix $A = \begin{pmatrix} x & 2 \\ 2 & 4 \end{pmatrix}$ does not have an inverse?
8. Find the conjugate of $Z_1 - Z_2$ if $Z_1 = 2 + 3i$ and $Z_2 = 5 + 3i$.
9. Find real x and y if $(x - iy)(3 + 5i)$ is the conjugate of $-6 - 24i$.
10. In FULafia sport competition, there are Faculties that play either Baseball or Football or both. The number of Faculties who play Baseball or Football or both are 6, 5 and 3 respectively. What will be the number of Faculties playing either cricket or badminton?

Submission Dateline: 12pm on Friday 25st July, 2023.

Submit in the classroom please

Dr. H. Ibrahim

Course Coordinator